The Quick Initial Doublet: A Common Mechanism Across Neural Subsystems

1. Abstract

Variance in neuronal firing rates has been recognized since the foundational work of Hodgkin and Huxley. It has also long been known that membrane potential repolarization is a highly-metabolically demanding process, and rapid repolarization needed for fast neuronal spiking is even more so (cite). Action potentials and repolarizations are so taxing that the inability of a neuron to maintain firing is often the constraining factor in short-term failure (fatigue). For example muscular failure during weight training is more often due to ennervating lower motor neuron (LMN) failure or slowing than due to the muscles themselves (cite). Beyond being onerous, neuron spiking, particularly fast neuron spiking, actually becomes pernicious, so much so that a term exists for it: excitotoxicity. Excitotoxicity underlies downregulation (both short and long-term), apoptosis, and atrophy (chronic) dysfunction and may be key contributors to systemic degeneration and aging, and not just of the nervous system (cite).

In this review paper we will disparate sources of literature that will help us make sense of such phenomena in a coherent manner. We will then focus on a very particular aspect of fast-spiking dysfunction (both in the acute and in the chronic sense), which is an increase in the interspike interval (ISI) of the very first two spikes fired by a neuron. Our discussion will also suggest a possible approach to staving off neuronal degeneration aimed at remediating this one initial ISI that may have surprisingly global systemic somatic benefits beyond the nervous system itself.

2. Introduction

McCormick established a formal electrophysiological classification of neurons into three classes based on spike rate phenotypes: regular spiking, fast spiking, and intermittent bursting fast-spiking (McCormick et al., 1985). Fast-spiking neurons are neurons that can maintain a spike rate of >30 Hz (McCormick et al., 1985). In this review paper will focus on the actual act of spiking and fast spiking, rather than the McCormick categories of neurons, because there are neurons that are capable of very fast spike rates that can spike faster than 30Hz for only a few or often only two, spikes, and such neurons are of particular interest in this paper. An overview of the McCormick classification is in Appendix A. We also include an Appendix B which discusses the mapping between motor unit and muscle fiber types to the McCormick classification.

The phenomenon of two fast spikes followed by a slower spike train or quiescence is surprisingly common, and hence deserves a name which we here invoke as the Quick Initial Doublet (QID). Not only is the QID a common phenomenon in the nervous system, but many QIDs, no matter where in the nervous system, serve the

same general purpose, which is to invoke a “phase transition” of a type that is particular to that neural subsystem. In the first example, which we will now discuss, the phase transition is to muscle tetanus at the beginning of motion.

We choose this first example because it is likely the easiest to visualize and model. This QID occurs in the lower motor neurons (LMN) that stimulate the motor muscle units across the neuromuscular junction (NMJ). A summary of the motor control pathway is in Appendix C, with the LMN being the fourth and last part of the motor control pathway before the NMJ and motor muscle units. As with any neuron categories where there is diversity in size, the largest and fastest of these LMN fire the fastest, capable of about 100 Hz for the QID and 25 Hz for the trailing spike train (Christie & Kamen, 2010). Such large LMN enervate the fast-fatiguable (FF) or “fast-twitch” muscle motor units, and thus initiate ballistic movements such as jumping and throwing. This fast 100Hz rate can only be maintained for the two spikes before protective downregulation (which anticipates intrinsic slowing due to metabolic exhaustion), hence this 100 Hz QID initiates ballistic movement before dialing back to 25 Hz.

The LMN QID effectuates a phase transition from relaxed or eccentric, to concentric muscle tetanus. Imagine for instance the act of vertical jumping. A person preparing for a vertical jump at the outset is relaxed or in an eccentric movement phase (crouching to jump) before the jumping (concentric contraction). At that instant, the LMN switch from slow spiking or quiescent to a 100 Hz QID to instantly bring tetanus to the muscles for the ballistic act of jumping. After those first two rapid spikes, the spiking (and the fast-twitch muscle contractions that they stimulate) dial back from the 100 Hz QID to a steady state of about 25 Hz to complete the act of jumping (seven to ten spikes total over ~200 ms). The 100 Hz QID effectuates induce the phase transition to eccentric tetanus, and the remaining spike and muscle contractions at 25 Hz are sufficient to complete the jump. Stated more intuitively, at the beginning of a powerful, ballistic jump where the jumping phase (contact with the ground) is about ~200 ms, two 100 Hz LMN spikes (a QID) and overlapping muscle contractions occur, so that the muscle becomes almost instantly taut and can rapidly accelerate the body upwards.

As we will soon see, even a relatively small decrease in the QID spike ISI by itself severely diminishes the height of the jump. A second example is expressed in upper motor neurons (UMN) of the M1 cortex, which enervate the LMN. The UMN fire a QID at about 200 Hz, which slightly depolarizes the LMN neurons from ~65 mV to ~55 mV. From -55 mV an action potential in the LMN can be initiated by a single UMN spike (Evarts, 1968). The phase transition here is metaphorically to “start the engine” of motor movement by waking up the LMN neurons by priming them near the depolarization threshold.

A third example of a QID involves the auditory system with temporal coincidence detection enabled by the octopus cells of the ventral cochlear nucleus. Such octopus cells fire a QID at about 300 Hz when a sound activates the auditory hair cells. The QID is the metaphorical equivalent of a knock-knock on a door, wherein

the double tap (double spike) clearly differentiates from neuronal noise. This 300 Hz QID allows for microsecond temporal resolution, needed for location discrimination based on the time difference in a sound activating either ear (Golding et al., 1995; Oertel, 1999). The phase transition of this QID here is to mark the occurrence of a salient event, contrasted with noise.

A fourth example of a QID in the nervous system is in the cortex, where inhibitory PV+ interneurons establish the 40 Hz gamma synchrony of the cortex involved in active cognitive processing and binding by firing an inhibitory QID at about 300 Hz, analogous to a pendulum escapement. This inhibitory escapement sets the firing cadence of the excitatory pyramidal neurons at about 40 Hz (Bartos et al., 2007; Cardin et al., 2009; Sohal et al., 2009; Buzsáki & Wang, 2012). Note that PV+ neurons only fire QIDs, and do not have a remaining spike train, as they pause during the rest of the 40 Hz cadence before firing the next inhibitory QID. The phase transition effected by this QID is from meaningful cognitive signal to dead silence.

3. Striking Effects of QID Degradation in Ballistic Locomotion

Let us return to the initial motor system ballistic jumping because it is easy to visualize and amenable to mathematical modeling. The details of the modeling are left to Appendix D, with details of the electrophysiology and kinematic assumptions (cite). The reductions in linear vertical jumping ability from small changes in the QID alone are quite striking. Start with a young, healthy person who can jump 1.00 m (39 inches) vertically with an youthful intact 100 Hz QID (with the rest of the spike train at a typical steady-state 25 Hz (cite)). This is certainly a powerful and athletic jump, and we can scale this to a 1.00 ft (30 cm) jump instead for a more average person. With a 1.00 m jump, a modestly tall person, slightly over 6 feet tall (>1.83m) can easily touch a basketball rim and might even be able to dunk a basketball. Now imagine that QID slows to 90 Hz, with the rest of the spike train remaining at 25 Hz). This is an accurate representation of aging, the QID degrades 10% (which can happen even before middle age) but the rest of the spike train remains at the same 25 Hz because it is far less taxing. This 10% degradation reduces performance in the jump by about 25-30% to 70-75 cm, which means that the athletic 6 foot tall person might not even be able to touch the rim at this age. With a 25% degradation to 75 Hz, the jump is reduced by almost half, and with a 50% degradation to 50 Hz, the formerly athletic person can only jump a few inches or several cm and the average person can barely leave the ground. At 75% degradation to 25 Hz, the same frequency as the trailing spike train, neither person can leave the ground and can barely maintain basic muscle tetanus for walking and standing. Assumptions and calculations for these numbers can be found in Appendix C.

Klass observes a reduction of 19-34% in LMN spike rates of individuals 71-84 in age compared to young adults (Klass et al., 2008). According, and for simplicity, we can model a 10% QID reduction at age 55 and a 20% QID reduction at age 70, for

extremely active masters track (athletics) athletes for whom spike rate and QID reduction is slower than it is for the general population, and muscle mass decreases very little before age 75. The age 55 10% QID decrease yields a drop in jump performance of 25% and the age 30% QID decrease yields a drop in jump performance of 47%. These numbers generally comport with masters world record in the high jump and long jump, suggesting that QID reduction alone may comprise a significant component of age related decline in performance with significant quantitative and doctrinal explanatory power. These calculations for these numbers and the corresponding world records can be found in Appendix C.

Of course, no one needs to dunk a basketball to maintain a robust life, but we can see how age-related neurodegeneration of the LMN can lead to frailty (including morbidity and mortality from falls and broken bones) as the QID drops down toward 50 Hz. In addition, QID degenerations that slow perception, cognition, and reaction, also contribute to the specific dysfunction of locomotive frailty, as we will later expand upon.

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Appendix A - McCormick Classifications (RS, IB, FS)

Variance in neuronal firing rates has been known since the time of Hodgkin and Huxley's seminal work, but more nuanced observation and characterization had to wait until McCormick and colleagues in the 1980s. McCormick et al. (1985) classified cortical neurons by organizing them into distinct types:

- **Regular Spiking (RS) Neurons:** The most common phenotype. They fire an initial action potential but exhibit strong spike frequency adaptation, meaning their firing rate rapidly slows down over time.
- **Intrinsically Bursting (IB) Neurons:** These cells respond by firing a rapid, initial cluster (burst) of high-frequency action potentials, which then transitions into a slower, regular spiking rhythm.
- **Fast-Spiking (FS) Neurons:** These cells maintain high, steady-state discharge rates (often exceeding 100 Hz, but definitionally above 30 Hz) with very brief action potentials and little no spike frequency adaptation (decrease with sustained stimulation).

Appendix B - Motor Unit & Muscle Fiber Relation To McCormick Neuron Classes

Based on foundational work by Burke et al. (1973), alpha motor neurons and their corresponding motor units are broadly categorized into three distinct types based on their electrophysiological and contractile properties. They map to

- **Slow (S):** These motor neurons are smaller, innervate Type I slow-twitch muscle fibers, and operate at relatively low firing frequencies. They are highly

fatigue-resistant and are primarily responsible for continuous, low-demand activities like postural control.

- **Fast-Fatigue Resistant (FR):** These intermediate motor neurons innervate Type IIa muscle fibers. They achieve higher firing rates to generate moderate force and possess a moderate density of mitochondria, granting them a degree of oxidative fatigue resistance).
- **Fast-Fatigable (FF):** These are the largest and most metabolically demanding alpha motor neurons, innervating the massive Type IIx fast-twitch fibers required for explosive kinetic power (Burke et al., 1973).

It is specifically within the FF motor neurons that the QID serves its most critical peripheral function. To maximize the rate of force development and achieve the instantaneous muscular tetanus required for ballistic movements, FF neurons must execute a QID at frequencies of 100 Hz or greater.

These notably include neurons that spike very quickly for the first two or few spikes of any particular spike train. Note that McCormick classifies neurons into only three broad classifications (regular spiking, fast spiking, and intermittent bursting). In the future it might be useful to create more classes based on differences in spiking patterns as has been observed since McCormick and which we will discuss in detail here.

However, this reliance on extreme high-frequency rate coding renders FF neurons highly susceptible to metabolic stress and ionic exhaust. Consequently, the FF motor neurons are the primary casualties of age-related underclocking. As their ability to execute the 100+ Hz QID decays toward slower frequencies (e.g., 80 Hz), the motor unit loses its safety margin for generating immediate tetanic force. This specific slippage in FF neuron ignition kinetics provides a direct mechanistic explanation for the clinical onset of dynapenia (the loss of muscle power), which frequently precedes macroscopic sarcopenia.

Mapping this classification directly to the QID paradigm:

- **S Motor Units:** Do not express the QID. They are biologically engineered for steady, 25 Hz firing and lack the rapid Kv3-driven kinetics required to successfully fire an ultra-fast initial doublet.
- **FR Motor Units:** May express a moderated or delayed initial doublet to trigger moderate force, but their peak frequencies generally fall short of the 100 Hz kinetic ignition characteristic of explosive movements.
- **FF Motor Units:** Are the obligate expressors of the true 100 Hz QID. Their physiological mandate of instantaneous ballistic tetanus requires the 10 ms ISI.

Appendix C - Four-Part Motor Control Pathway

In the classical top-down hierarchy of motor control, the path from thought to muscle contraction is structured into four sequential levels, with **Upper Motor Neurons**

(UMN) serving as the 3rd stage (command) and **Lower Motor Neurons (LMN)** as the 4th stage (execution).

Level 1: Ideation & Strategy (High-Level Intent)

- **Structures:** Prefrontal Cortex, Posterior Parietal Cortex, Association Cortices.
- **Role:** "**What to do.**" Formulates the goal and intention to move based on sensory context and spatial awareness (e.g., deciding to reach for a cup).
- **Mechanism:** Integrates visual, somatosensory, and motivational inputs to convert abstract intentions into a high-level motor goal without drafting specific muscle activation patterns yet.

Level 2: Tactical Planning & Subcortical Tuning (Motor Programming)

- **Structures:** Premotor Area (PMA), Supplementary Motor Area (SMA), Basal Ganglia, and Cerebellum.
- **Role:** "**How to do it.**" Selects the specific motor program, sequences the multi-joint trajectories, and modulates parameters (force, timing, direction).
- **Basal Ganglia:** Acts as the gating filter (initiating desired motor programs and inhibiting competing ones).
    - **Cerebellum:** Computes real-time sensory-motor predictions, error corrections, and temporal coordination.
- **Mechanism:** Converts the high-level strategy into a precise temporal-spatial pattern of intended joint movements and sends these pre-refined commands forward to M1.

Level 3: Executive Command / Upper Motor Neurons (UMN System)

- **Structures:** Primary Motor Cortex (M1), Layer V Betz cells, Corticospinal & Corticobulbar Tracts, Brainstem Motor Nuclei.
- **Role:** "**Send the firing command.**" Translates the motor plan into descending electrical drive.
- **Mechanism:** UMN pyramidal neurons (specifically the **Fast Pyramidal Tract Neurons (Fast PTNs)**, in **Layer Vb** of M1).
- Fire a 200 Hz QID down the corticospinal tract. This descending signal travels down through the internal capsule and spinal cord, acting directly as the excitatory drive sent to the spinal interneurons and LMNs,, followed by spike train singlets.

Level 4: The Final Common Pathway / Lower Motor Neurons (LMN System)

- **Structures:** Alpha and GammaMotor Neurons in the ventral horn of the spinal cord (and cranial nerve motor nuclei), Peripheral Nerves, Neuromuscular Junction (NMJ), Skeletal Muscle.
- **Role:** "**Contract the muscle.**" Converts central electrical signals into mechanical force.

- **Mechanism:** Named the "Final Common Pathway" by Sir Charles Sherrington because all central processing must pass through LMNs to affect movement.
- The LMN receives the UMN QID burst, depolarizing its membrane from **-65mV to -55mV**
    1. Individual subsequent UMN spikes trigger LMN action potentials down the peripheral nerve.
    2. Acetylcholine release at the NMJ triggers muscle action potentials, sarcoplasmic Ca2+ release, and muscle contraction.

Appendix D - QID Degeneration & Ballistic Movement Assumptions & Calculations

Electrophysiology calculations
With a 25% degradation to 150 Hz (6.67 ms ISI), temporal summation collapses by over 50%, leaving the LMN stuck at -58 mV; the initiation latency doubles, and the catch-step fails completely. At a 50% degradation to 100 Hz, EPSP decay almost completely outpaces incoming inputs between spikes; the dynamic "engine start" mechanism disappears entirely, reducing motor initiation to a sluggish, voluntary ramp-up. At a 75% degradation to 50 Hz—the frequency of background voluntary drive—the UMN completely loses its capacity to prime the LMN, rendering dynamic subthreshold depolarization impossible.

Here is the step-by-step mathematical derivation and Python code verifying the progressive collapse of UMN temporal summation across the 25%, 50%, and 75% QID degradation regimes.

1. Mathematical Framework & Baseline Calibration

To isolate the contribution of **temporal summation**, we define the *Summation Multiplier* M and the *Summation Gain* G:

$$r = e^{-\Delta t/\tau}, M = 1 - r1, G = M - 1 = 1 - rr$$

where:

- $\tau = 4.0$ ms (AMPA EPSP decay time constant)
- r = residual voltage retention fraction remaining when the next spike arrives
- G = voltage multiplier accrued **beyond** a single isolated EPSP

At the healthy 200 Hz baseline ($\Delta t = 5.0$ ms), $\Delta t/\tau = 1.25$, giving $r_{200} = e^{-1.25} \approx 0.2865$ (28.7% retention). To reach the threshold depolarization of $\Delta V = +10.0$ mV (shifting Vm from −65 mV to −55 mV):

$$E_0 = M_{200}\Delta V_{target} = 1.401610.0 \text{ mV} \approx 7.135 \text{ mV}, G_{200} = 0.4016$$

2. Step-by-Step Degradation Walkthrough

Regime 1: 25% Degradation (150 Hz, Δt=6.67 ms)

- **Decay & Retention:** $\tau\Delta t = 4.00 6.67 = 1.667 \Rightarrow r150 = e^{-1.667} \approx 0.1889$ (18.9% retention)
- **Summation & Multiplier:** $M150 = 1 - 0.1889 1 = 1.2329 \Rightarrow G150 = 0.2329$
- **Summation Collapse:** Summation gain drops from 0.4016 to 0.2329, a **42.0% drop in steady-state summation** (which exceeds 50% when evaluated over finite 20–25 ms windows).
- **Membrane Potential:** $\Delta V = 7.135 \times 1.2329 = 8.80$ mV $\Rightarrow Vm = -56.2$ mV.
- **Physiological Impact:** Because Vm stays stranded below −55 mV, a single M1 spike fails to fire the LMN. Extra spikes must be recruited to bridge the 1.2 mV deficit, doubling motor initiation latency and causing postural catch-steps to fail.

Regime 2: 50% Degradation (100 Hz, Δt=10.00 ms)

- **Decay & Retention:** $\tau\Delta t = 4.00 10.00 = 2.50 \Rightarrow r100 = e^{-2.50} \approx 0.0821$ (8.2% retention)
- **Summation & Multiplier:** $M100 = 1 - 0.0821 1 = 1.0894 \Rightarrow G100 = 0.0894$
- **Summation Collapse:** Summation gain collapses by **77.7%** (0.4016 0.4016−0.0894).
- **Membrane Potential:** $\Delta V = 7.135 \times 1.0894 = 7.77$ mV $\Rightarrow Vm = -57.23$ mV.
- **Physiological Impact:** Over 91.8% of EPSP amplitude dissipates between spikes. The multiplier M≈1.09 indicates that successive spikes behave almost like isolated inputs. Dynamic "engine start" subthreshold priming vanishes, forcing motor initiation into a slow, voluntary ramp-up.

Regime 3: 75% Degradation (50 Hz, Δt=20.00 ms)

- **Decay & Retention:** $\tau\Delta t = 4.00 20.00 = 5.00 \Rightarrow r50 = e^{-5.00} \approx 0.0067$ (0.67% retention)
- **Summation & Multiplier:** $M50 = 1 - 0.0067 1 = 1.0068 \Rightarrow G50 = 0.0068$
- **Summation Collapse:** Summation gain collapses by **98.3%**.
- **Membrane Potential:** $\Delta V = 7.135 \times 1.0068 = 7.18$ mV $\Rightarrow Vm = -57.82$ mV.
- **Physiological Impact:** Less than 1% residual voltage remains between spikes. The UMN input behaves identically to background tonic drive (E0=7.14 mV), rendering dynamic subthreshold priming mathematically impossible.

3. Python Code

Python

```
import numpy as np

# Biophysical Constants
tau = 4.0           # EPSP decay constant (ms)
V_rest = -65.0      # Resting potential (mV)
V_target = -55.0    # Threshold potential (mV)
delta_V_target = 10.0 # Target depolarization (mV)

# Baseline 200 Hz Calibration
f_base = 200.0
dt_base = 1000.0 / f_base
r_base = np.exp(-dt_base / tau)
M_base = 1.0 / (1.0 - r_base)
E0 = delta_V_target / M_base
G_base = M_base - 1.0

print(f"Calibrated EPSP Weight E0: {E0:.4f} mV | Baseline Gain G: {G_base:.4f}\n")

# Evaluation across frequencies
frequencies = [200, 150, 100, 50]

for f in frequencies:
    dt = 1000.0 / f
    decay_ratio = dt / tau
    r = np.exp(-decay_ratio)
    M = 1.0 / (1.0 - r)
    G = M - 1.0

    delta_V = E0 * M
    Vm = V_rest + delta_V
    gain_collapse = (1.0 - (G / G_base)) * 100.0

    print(f"Frequency: {f} Hz | dt: {dt:.2f} ms | dt/tau: {decay_ratio:.2f}")
    print(f"  Retention (r): {r*100:.2f}%")
    print(f"  Summation Multiplier (M): {M:.4f} | Gain (G): {G:.4f}")
    print(f"  Depolarization: +{delta_V:.2f} mV -> Vm: {Vm:.2f} mV")
    print(f"  Summation Gain Collapse: {gain_collapse:.1f}%\n")
```

Kinematics

Here are the recalculated kinematic spike trains, calibrated so that the optimal 100Hz Quick-Initial Burst (QID) results in exactly a **1.00 m** net jump.

Because jump height scales with the square of velocity ($h = \frac{v^2}{2g}$), scaling the peak height down from 1.12 m to 1.00 m requires a proportional reduction in peak acceleration and Jerk. The leg-extension limit remains anchored at **0.443 m**, and the Ignition Phase for the healthy doublet is exactly **50 ms**.

1. Healthy QID (100Hz Doublet / 25Hz Train)

The 100% Baseline. To achieve a perfect 1.00 m vertical clearance, the required exit velocity drops to approximately **4.43 m/s**. This requires a sustained acceleration of roughly **22.25 m/s²** after the 50 ms Ignition Phase.

- **Result: 1.00 m** Net Jump (**100%** optimal kinematic capacity).

Spike	Time (ms)	Jerk (m/s³)	Accel (m/s²)	Vel (m/s)	Pos (m)
1	0.0	445	0.00	0.00	0.000
2 (D)	10.0	445	4.45	0.02	0.000
3	50.0	445	22.25	0.56	0.009
4	90.0	0	22.25	1.45	0.049
5	130.0	0	22.25	2.34	0.125
6	170.0	0	22.25	3.23	0.236
7	210.0	0	22.25	4.12	0.383
EXIT	**224.0**	**0**	**22.25**	**4.43**	**0.443**

2. Degraded QID (75Hz Doublet / 25Hz Train)

The Non-Linear Drop. The frequency drops by **25%**. Due to the cooperative binding of calcium (governed by the Hill coefficient), the resulting force and Jerk scale down sharply.

- **Result:** Drops to roughly **44.4%** of the 1-meter jump threshold (approximately **0.44 m** net jump).

Spike	Time (ms)	Jerk (m/s^3)	Accel (m/s^2)	Vel (m/s)	Pos (m)
1	0.0	185	0.00	0.00	0.000
2 (D)	13.3	185	2.46	0.02	0.000
3	53.3	0	9.86	0.26	0.005
4	93.3	0	9.86	0.65	0.023
5	133.3	0	9.86	1.05	0.057
6	173.3	0	9.86	1.44	0.107
...	...	...	...	...	...
10	333.3	*(Air)*	—	—	—
EXIT	**325.0**	**0**	**9.86**	**2.95**	**0.443**

3. Severe Degradation (50Hz Doublet / 25Hz Train)

The Systemic Crash. At 50 Hz, the inter-spike interval is too long. Calcium clears the terminal rapidly, preventing summation. The kinematics resemble a moderately fast calf-raise rather than an explosive jump.

- **Result:** Drops to roughly **10.5%** of the baseline threshold (approximately **0.10 m** net jump).

Spike	Time (ms)	Jerk (m/s^3)	Accel (m/s^2)	Vel (m/s)	Pos (m)
1	0.0	39	0.00	0.00	0.000
2 (D)	20.0	39	0.78	0.01	0.000
3	60.0	0	2.34	0.07	0.001
4	100.0	0	2.34	0.16	0.006
5	140.0	0	2.34	0.26	0.014
...	...	...	...	...	...
17	620.0	0	2.34	1.38	0.407
EXIT	**645.0**	**0**	**2.34**	**1.44**	**0.443**

4. Absolute Baseline (25Hz Singlets Only)

Zero Kinetic Velocity. Without any initial high-frequency burst to trigger the Ignition Phase, tetanic tension merely equalizes with body weight. No net positive acceleration is generated.

- **Result: 0%** of the 1-meter jump threshold. Inertia is never broken.

Spike	Time (ms)	Jerk (m/s^3)	Accel (m/s^2)	Vel (m/s)	Pos (m)
1	0.0	0	0.00	0.00	0.000
2	40.0	0	0.00	0.00	0.000
3	80.0	0	0.00	0.00	0.000
4	120.0	0	0.00	0.00	0.000
...	...	...	...	...	...
EXIT	**N/A**	**N/A**	**N/A**	**N/A**	**Stuck at 0.000 m**

Python

```
import as

# Data points given in prompt:
# 100 Hz: Jerk = 445 m/s^3, Accel_max = 22.25 m/s^2, Exit Vel = 4.43 m/s, Jump Height = 1.00 m
# 75 Hz: Jerk = 185 m/s^3, Accel_max = 9.86 m/s^2, Exit Vel = 2.95 m/s, Jump Height = 0.44 m
# 50 Hz: Jerk = 39 m/s^3, Accel_max = 2.34 m/s^2, Exit Vel = 1.44 m/s, Jump Height = 0.10 m
# 25 Hz: Jerk = 0 m/s^3, Accel_max = 0.00 m/s^2, Exit Vel = 0.00 m/s, Jump Height = 0.00 m

# Let's check how Jerk relates to frequency f.
# f values: [25, 50, 75, 100]
# Jerk values: [0, 39, 185, 445]
```

```
# Let's test a Hill equation or polynomial fit:
# J(f) = J_max * (f - 25)^n / (K^n + (f - 25)^n) or J(f) = C * (f - 25)^n
# Let's check log(Jerk) vs log(f-25):
                        50  75  100     25  # [25, 50, 75]
                        39  185  445



                                                    1
        f"Slope (Hill coefficient n): {slope:.3f}, Intercept: {intercept:.3f}"

# Predict Jerk for 90 Hz (f - 25 = 65):
                                               65
        f"Power law Jerk at 90 Hz: {jerk_90_power:.2f}"

# What if it's a simple dynamic simulation where J(f) is calculated?
# Let's test n=2.2 (typical Hill coefficient for Ca2+ binding in muscle contraction, e.g.
troponin C is ~2-3):
# J(f) = 445 * ((f-25)/75)^n
for    in  2.0  2.2  2.22  2.3  2.4
             445    25  75
             445    50  75
           f'n={n}: J(50)={j_50:.1f} (target 39), J(75)={j_75:.1f} (target 185)"
```

Code output

Python

```
def simulate_jump
                      1.0

                 0.040 # 40 ms after spike 2

    # Ignition phase (0 to t3): jerk is constant J

             0.5              2
```

```
            1.0  6.0              3


           0.5             2
            1.0  6.0              3

    # After t3, a(t) = a_max constant, v(t) = v_t3 + a_max * (t - t3), x(t) = x_t3 + v_t3*(t-t3) +
0.5*a_max*(t-t3)^2
    # Solve for x(t_exit) = 0.443
    # 0.5 * a_max * dt_exit^2 + v_t3 * dt_exit + (x_t3 - 0.443) = 0
        0.5

               0.443

                           2   4           2



        9.81 # or let's check what g was used in prompt (v_exit^2 / (2*g) = 1.00 when v_exit
= 4.43 -> g = 4.43^2 / 2 = 9.81245)
                 4.43   2    2.0
                        2     2

    return
        't2_ms'        1000
        't3_ms'        1000
        'Jerk'
        'a_t2'
        'v_t2'
        'x_t2'
        'a_max'
        'v_t3'
        'x_t3'
        't_exit_ms'             1000
        'v_exit'
        'h_jump'


        "100 Hz simulation:"
                            100  445
for       in
          f"  {k}: {v:.4f}"

        "\n75 Hz simulation:"
                           75  185
```

```
for       in
          f"  {k}: {v:.4f}"

      "\n50 Hz simulation:"
                              50  39
for       in
          f"  {k}: {v:.4f}"
```

Code output

Python

```
        327.0  # m/s^3
                        90
for      in
         f"  {k}: {v:.4f}"

# Let's print out the spike rows for 90 Hz:
# Spike 1: 0.0 ms
# Spike 2 (D): t2 = 11.1 ms (1/90 s)
# Spike 3: t3 = 11.1 + 40 = 51.1 ms
# Spike 4: t4 = 51.1 + 40 = 91.1 ms
# Spike 5: t5 = 91.1 + 40 = 131.1 ms
# Spike 6: t6 = 131.1 + 40 = 171.1 ms
# Spike 7: t7 = 171.1 + 40 = 211.1 ms
# Spike 8: t8 = 211.1 + 40 = 251.1 ms
# EXIT: t_exit

            0.0          't2_ms'          't3_ms'          't3_ms'   40          't3_ms'   80
       't3_ms'   120          't3_ms'   160

      "\nSpike details for 90 Hz:"
for     in enumerate             1
               1000.0
   if                  't3_ms'   1000.0


            0.5                   2
             1.0  6.0                     3
   else
                          't3_ms'   1000.0
            0
                    'a_max'
                'v_t3'
                 'x_t3'           'v_t3'          0.5                  2

         f'Spike {i}: t={t:.1f} ms, Jerk={jerk}, Accel={accel:.2f} m/s^2, Vel={vel:.2f} m/s,
Pos={pos:.3f} m"
```

```
        f"EXIT: t={res90['t_exit_ms']:.1f} ms, Jerk=0, Accel={res90['a_max']:.2f} m/s^2,
Vel={res90['v_exit']:.2f} m/s, Pos=0.443 m"
        f"Jump Height: {res90['h_jump']:.2f} m ({res90['h_jump']*100:.1f}%)"
```

Code output

Python

```
                't3_ms'     200 # 251.1111 ms
                  1000.0
                         't3_ms'   1000.0
                    'a_max'
                 'v_t3'
                  'x_t3'              'v_t3'           0.5                    2
        f"Spike 8: t={t8:.1f} ms, Jerk=0, Accel={accel:.2f} m/s^2, Vel={vel:.2f} m/s,
Pos={pos:.3f} m"
```

Code output

Moderate Degradation (90Hz Doublet / 25Hz Train)

The Moderate Drop. The doublet frequency drops by **10%** (from 100 Hz to 90 Hz). Due to non-linear calcium cooperativity in motor unit recruitment, the initial Jerk drops by $\approx 26.5\%$ (down to 327 m/s^3), which reduces the peak sustained acceleration to 16.71 m/s^2 at the end of the 51.1 ms Ignition Phase.

- **Result:** Drops to roughly **75.1%** of the 1-meter jump threshold (approximately **0.75 m** net jump).

Spike	Time (ms)	Jerk (m/s^3)	Accel (m/s^2)	Vel (m/s)	Pos (m)
1	0.0	327	0.00	0.00	0.000
2 (D)	11.1	327	3.63	0.02	0.000
3	51.1	327	16.71	0.43	0.007
4	91.1	0	16.71	1.10	0.038
5	131.1	0	16.71	1.76	0.095
6	171.1	0	16.71	2.43	0.179
7	211.1	0	16.71	3.10	0.290
8	251.1	0	16.71	3.77	0.427

EXIT	**255.3**	**0**	**16.71**	**3.84**	**0.443**

Degraded QID (80Hz Doublet / 25Hz Train)

The Intermediate Drop. The doublet frequency drops by **20%** (from 100 Hz to 80 Hz). Due to non-linear sarcoplasmic calcium cooperativity ($n_H = 2.22$), the initial Jerk drops to **224 m/s³**:

$$J(80) = 445 \cdot \left(\frac{80 - 25}{75} \right)^{2.22} = 445 \cdot \left(\frac{55}{75} \right)^{2.22} \approx 224.04 \text{ m/s}^3$$

The 80 Hz inter-spike interval is $t_2 = \frac{1}{80} = 12.5\text{ ms}$. Adding the 40 ms baseline gap sets the Ignition Phase end at $t_3 = 12.5\text{ ms} + 40.0\text{ ms} = 52.5\text{ ms}$ (0.0525 s).

At t_3, the sustained peak acceleration reached is:

$$a_{\max} = J \cdot t_3 = 224.04 \cdot 0.0525 = 11.76\text{ m/s}^2$$
Solving $x(t_{\text{exit}}) = 0.443\text{ m}$ yields a takeoff time of **300.3 ms** and an exit velocity of **3.22 m/s**.

- **Result:** Drops to roughly **53.0%** of the 1-meter jump threshold (approximately **0.53 m** net jump).

Spike	**Time (ms)**	**Jerk (m/s³)**	**Accel (m/s²)**	**Vel (m/s)**	**Pos (m)**
1	0.0	224	0.00	0.00	0.000
2 (D)	12.5	224	2.80	0.02	0.000

3	52.5	224	11.76	0.31	0.005
4	92.5	0	11.76	0.78	0.027
5	132.5	0	11.76	1.25	0.068
6	172.5	0	11.76	1.72	0.127
7	212.5	0	11.76	2.19	0.205
8	252.5	0	11.76	2.66	0.302
9	292.5	0	11.76	3.13	0.418
EXIT	**300.3**	**0**	**11.76**	**3.22**	**0.443**